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Recovering Additional Satellite Telemetry from Archived Audio with a Resumable Multi-Hypothesis Receiver

AUTHOR NAMES TO BE SUPPLIED

Affiliations and author identifiers to be supplied before submission.

Corresponding author and contact details to be supplied.

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ABSTRACT Archived satellite audio provides an opportunity to recover telemetry without repeating a radio observation. We describe Telemetry Yield, a Rust implementation of a resumable multi-hypothesis receiver for post-frequency-demodulation recordings. Its progressive portfolio combines complementary frontends, timing hypotheses, guarded channel estimation from separately received packets, and sequence detection. An additional experimental receiver evaluates residual prediction and codec-derived reliability proxies. A frozen interim comparison uses 189 completed observations from a planned cohort of 266 historical CANVAS recordings, with identical numerical audio supplied to Dire Wolf, gr-satellites, and the native receivers. The progressive receiver recovers 4541 distinct AX.25 unnumbered-information frames counted within observations, compared with 2998 in the union of the two external receivers. It adds 1547 frames while missing four reference frames, giving a net increase of 51.5 percent. The experimental codec-enabled and codec-disabled receivers produce identical packet sets in every completed observation, providing no incremental packet-yield evidence for codec conditioning in this snapshot. These results support further evaluation of complementary offline decoding, not universal receiver superiority. The study remains incomplete, reuses a historically exposed cohort after a performance-only restart, and does not establish spacecraft authenticity, broad modulation coverage, or deployment readiness.

INDEX TERMS Archived radio audio, channel estimation, packet radio, satellite telemetry, sequence detection, software-defined radio.

I. INTRODUCTION

ARCHIVED satellite observations contain information that a particular receiver configuration may fail to convert into accepted packets. Reprocessing the same recording can therefore be useful even when the original observation has already ended. This is a receiver-side recovery problem: it neither increases the transmitted information nor reconstructs information that a lossy representation has irreversibly removed.

This paper describes an implemented offline system and a preliminary historical evaluation. The central question is whether a bounded, complementary decoding portfolio can recover additional validated packets from the same numerical audio used by established receivers. A second question concerns the value of residual prediction and codec-derived

reliability information. The two questions involve different receiver configurations and must not be conflated.

The primary receiver, denoted P, is the checkpointable `progressive_v3` portfolio. The separate experimental receiver, I2, includes residual-predictive and codec-conditioned lanes. Its control I1 is `innovation_v1_no_codec`. Dire Wolf and gr-satellites are independent replay comparators, not interchangeable names for the complete native SatNOGS receiving chain. The input recordings originate from SatNOGS; the reported reference counts are generated by the controlled replays in this study, not copied from its public packet database.

The proposed contribution is a reproducible receiver architecture combining timing diversity, disjoint cross-packet channel-model transfer, additive sequence-detection branches, and resumable execution. No claim is made that timing

recovery, maximum-likelihood sequence estimation (MLSE), autoregressive prediction, or weighted least squares is individually new. The engineering combination and its empirical behavior are the subject of investigation. Establishing priority for a narrowly defined new algorithm requires a broader prior-art review and matched component ablations.

II. RELATED METHODS AND BENCHMARK CONTEXT

MLSE for channels with intersymbol interference is established theory [1]. Timing recovery likewise has longstanding methods, including Gardner’s timing-error detector [2]. Estimation in autoregressive noise is also well studied [3]. The present implementation uses these families as building blocks; an exact trellis search under an approximate fitted channel is not an assertion that the physical satellite receiver is globally optimal.

Dire Wolf’s WA8LMF results document is valuable methodological context [4]. Its January 2019 revision compiles measurements made by different operators and explicitly warns about track selection, test conditions, software age, and the interpretation of small differences. The WA8LMF source describes a standard 1200-baud packet-radio test and distinguishes discriminator audio from de-emphasized audio [5]. These distinctions matter because frontend processing changes the signal presented to a decoder.

We do not compare our CANVAS packet totals numerically with that compilation. The present study uses 9600-baud satellite audio, a different corpus, and per-observation exact-packet deduplication. No Telemetry Yield result on the WA8LMF test medium is reported here. A future controlled WA8LMF experiment must pin the medium version and track hashes, implement and qualify the appropriate 1200-baud audio path, and replay the identical samples through explicitly configured receivers. Version-specific raw packet totals from unrelated tests are not a common performance scale.

The gr-satellites command-line documentation distinguishes complex-input operation from audio input after frequency demodulation [6]. Our replay is in the latter domain. Consequently, this study does not measure coherent acquisition, raw in-phase/quadrature (IQ) performance, RF sensitivity in dBm, or a change in required signal-to-noise ratio.

III. PROGRESSIVE RECEIVER ARCHITECTURE

A. INPUT CONTRACT AND BASELINE PORTFOLIO

The evaluated study configuration accepts mono post-FM audio representing 9600-baud GMSK/G3RUH AX.25 traffic. Recordings are processed in 6-s windows with a 3-s hop. Overlap can yield repeated packet detections, which are removed when forming the observation-level packet set.

Two complementary frontends are retained: a legacy finite-impulse-response/DC-removal path and a square-pulse path with four-boxcar DC rejection and causal root-mean-square normalization. Each frontend evaluates 160 fixed timing hypotheses: 32 phases at each of five clock errors, -500 , -100 , 0 , 100 , and 500 parts per million. Sixteen Gardner

configurations per frontend combine two loop bandwidths (0.02 and 0.06) with eight initial phases.

The quick prefix comprises eight waveform-ranked fixed timings and two Gardner configurations per frontend, or 20 attempts per window. The remaining baseline comprises 332 attempts, giving 352 baseline attempts per window before supplemental sequence-detection work. Ranking uses the observed waveform, not reference packet contents. The finite search bank is intentionally retained; “quick” denotes a scheduling budget rather than a claim of identical yield after fewer hypotheses.

B. GUARDED CROSS-PACKET CHANNEL ESTIMATION

Native baseline detections are accepted only after a received frame-check sequence (FCS) check and strict AX.25 unnumbered-information (UI) structure checks. Plain non-return-to-zero inverted (NRZI) and NRZI/G3RUH hypotheses preserve the physical symbol coordinates needed for channel training. An accepted packet supplies a candidate training span; archive payloads and external-decoder packet bytes are never supplied to the receiver as training targets.

For binary symbols $x_n \in \{-1, +1\}$, a local symbol-rate model is

$$y_n = b + h_{-1}x_{n-1} + h_0x_n + h_{+1}x_{n+1} + e_n. \quad (1)$$

The implementation estimates the centered three-tap response and bias using ridge-regularized fitting, at least 128 distinct guarded training symbols, and channel-quality gates. This is an effective post-FM audio model, not a full physical model of a Gaussian minimum-shift-keying RF channel.

Source and target windows must use the same frontend, be within 60 s, and remain disjoint with at least a 1-s guard. The nearest eligible source is selected deterministically. Supplemental timing tests combine eight transferred-clock phase hypotheses and eight locally preferred hypotheses, removing exact duplicates. A source packet can therefore help model an adjacent difficult interval without fitting that target interval to a known desired payload. Temporal separation reduces direct training leakage but does not make neighboring windows statistically independent.

C. SEQUENCE DETECTION AND ADDITIVE BRANCHES

Given a fitted channel and a trial gain g , the white-residual lane minimizes

$$\hat{x}_g = \arg \min_{x \in \{-1, +1\}^N} \sum_n \left[y_n - g \left(b + \sum_{k=-1}^1 h_k x_{n+k} \right) \right]^2. \quad (2)$$

A deterministic four-state Viterbi implementation uses free initial/final states and full traceback. It additionally evaluates the configured gain factors 0.75, 1, and 1.5. MLSE here concerns symbol estimation in intersymbol interference; it is not convolutional error-control decoding. Exact optimization of the chosen metric does not remove channel-model mismatch.

A blind branch uses five bounded initial seeds and at most five alternating sequence-estimation/least-squares iterations.

A disjoint waveform-validation gate rejects poor models. A low held-out waveform error is not in itself evidence of a genuine satellite transmission. A further branch blends up to 16 compatible local anchor models using reliability and time-distance weights, rejecting inconsistent response shapes. This is local model pooling, not a general time-varying channel estimator or multi-station signal combination.

The baseline, nearest-anchor, blind, and multi-anchor packet outputs are combined by set union. Early supplemental work uses an immutable quick-prefix anchor generation; final supplemental work uses the completed baseline anchor generation. Newly recovered supplemental packets do not recursively become new training anchors. No CRC-guided bit repair is performed. Retaining the native baseline prevents a supplemental branch from deleting its packets, but does not guarantee that P contains every external receiver's packet.

D. ANYTIME EXECUTION AND RESOURCE USE

The quick, deep, and full modes change the execution budget of one finite task bank. Default quick and deep budgets are 3000 and 60000 ms. Preparation, hashing, and checkpoint verification consume this budget, but scheduling and cleanup overhead can exceed the nominal deadline; it is not a hard real-time guarantee. A later invocation can resume deeper work from an existing compatible checkpoint without repeating committed tasks.

Manifests bind the source, numerical input, executable, and policy. Immutable tasks, atomic checkpoints, and process locks protect restart consistency. A cached frontend/timing representation and reusable trellis buffers reduce redundant work while preserving floating-point arithmetic and the search bank. Independent work can execute in parallel. Cache size and thread count are resource settings, not an authorization to silently prune hypotheses. Incomplete work is explicitly different from a completed decode with zero packets.

IV. SEPARATE RESIDUAL AND CODEC EXPERIMENT

The I2 executable preserves an adaptive receiver baseline and adds matched-white, residual-innovations, codec-conditioned, and pooled-codec lanes. It is not currently a checkpointable task in P, and its retained baseline is not P's complete blind/multi-anchor portfolio. This architectural difference is important when interpreting the results.

For residuals e_n from (1), a candidate innovations metric is

$$J(x) = \sum_n \frac{\left(e_n - \sum_{j=1}^p a_j e_{n-j}\right)^2}{v_n}. \quad (3)$$

Here $p \in \{0, 1, 2\}$ and $v_n > 0$. Bounded stable autoregressive candidates are learned from source residuals with disjoint validation and a white-noise fallback. Constant-weight order zero delegates to the original detector to preserve its arithmetic. Prediction/whitening and sequence detection are established ideas; their use alone does not establish novelty.

Codec conditioning uses proxies derived from the original Vorbis stream: logarithmic encoded-packet bits per decoded

sample and adjacent decoded-frame length transitions. These features are calibrated against residual innovations with a held-out likelihood gate, rather than directly optimized for successful packet counts. Sample alignment follows cumulative decoded-frame lengths. The proxies are neither known quantization errors nor a complete parsed codec state.

Pooled calibration requires at least two disjoint source windows with distinct native received frames. Source identity, frontend identity, target separation, and training-only normalization remain constrained. Rejected calibration disables the optional weighted lane instead of invalidating the observation. The plausible research hypothesis is that such proxies may help select reliability weights. The interim packet results below do not support an incremental benefit for these lanes.

V. HISTORICAL EVALUATION PROTOCOL

A. COHORT AND EVALUATION EXPOSURE

The planned cohort comprises 266 historical CANVAS observations from 15 UTC dates and 161 ground-station identifiers. It was amended from 500 selected observations by excluding 234 under the recorded prior-exposure/proximity rule with a 120-min same-satellite guard. It is a selected single-mission cohort, not a representative sample of all SatNOGS modulation families or all observation outcomes.

The optimized executable campaign is a disclosed repeat evaluation after a performance-only restart. Earlier partial results had been inspected. The executable/build and report-reader changes did not intentionally alter receiver thresholds, cohort selection, or hypothesis banks, but re-evaluation does not restore unseen-holdout status. The archived protocol and exposure record are part of the evidence bundle.

The snapshot cutoff is September 12, 2026, 22:52:19 UTC. Exactly 189 observations have completed all five arms; 77 planned observations have no complete five-arm result in this snapshot. They are not assigned zero packets. The interim complete-case subset is not randomized by completion, so runtime-dependent selection can bias early estimates. No final confidence interval or significance declaration is inferred from it.

B. COMMON INPUTS AND COMPARATORS

FFmpeg converts each original recording once into a common signed 16-bit PCM WAV, retaining its sample rate and channels. Each decoder receives identical numerical WAV content. This explicit benchmark conversion differs from the native progressive CLI's usual floating-point OGG conversion. I2 additionally receives codec side information from the original OGG; this extra information is disclosed, not silently treated as identical input information.

Dire Wolf runs `atest -B 9600 -F 0 -h`, with bit repair disabled. The gr-satellites profile explicitly specifies FSK at 9600 baud and AX.25 G3RUH framing; it is a frozen generic link profile, not a claim of complete mission-specific application decoding. Network telemetry submission is disabled. Binary identities, profile content, arguments, and

selected dependency hashes are preserved in the bundle. Those hashes do not constitute a complete hermetic system image.

The campaign uses four observation workers, two threads per native arm, and a 900-s external arm cap. P receives an 880000-ms internal budget and must finish its required bank to count as complete. Arm order rotates across observations. Individual wall times are measured on a shared host; conversion is outside each arm’s timed process, and parallel CPU use differs between programs.

C. PACKET IDENTITY, VERIFICATION, AND METRICS

Let P_i , D_i , and G_i be the exact FCS-stripped UI packet-byte sets from P, Dire Wolf, and gr-satellites for observation i . Define $B_i = D_i \cup G_i$. A repeated packet counts once within an observation; the same bytes in a different observation count again. Thus “unique” does not mean globally new telemetry or distinct spacecraft measurements.

$$A = \sum_i |P_i \setminus B_i|, \quad L = \sum_i |B_i \setminus P_i|, \quad (4)$$

$$\Delta = A - L, \quad R = \frac{\sum_i |P_i| - \sum_i |B_i|}{\sum_i |B_i|}. \quad (5)$$

R is undefined for a zero reference denominator. Added packets, lost packets, net change, and observations with any addition are separately reported. This avoids hiding regressions behind a larger aggregate count.

Native outputs retain the received FCS and record independent bitwise rechecking. External outputs strip it and rely on each external decoder’s internal FCS check, followed by structural and exact-byte comparison. This evidence asymmetry must remain explicit. A valid FCS is not spacecraft authentication, and many correlated hypotheses complicate simple random-error arguments. The final artifact-level audit and dedicated false-positive controls remain prerequisites to a submission-quality claim.

D. CONFIRMED-SIGNAL SUBGROUP

The user-requested subgroup is frozen metadata with `waterfall_status=with-signal`, not a condition based on our packet recovery. The full cohort has 24 such labels, three without-signal labels, and 239 unknown labels. Sixteen of the 24 signal-labelled observations are complete at the cutoff. This subgroup was requested after initial results and is exploratory. A visible waveform does not prove the intended spacecraft identity, a supported modulation, or recoverable packet contents; unknown labels must not be interpreted as signal absence.

VI. PRELIMINARY RESULTS

Table 1 summarizes 87619.7 s (24.34 h) of recorded audio. P adds 1547 packets relative to B and misses four, yielding a net 1543 packets or 51.47%. There is at least one addition in 107 of 189 observations (56.6%). This is neither a packet success probability nor a fraction of all transmitted packets recovered: the transmitted ground truth is unavailable.

TABLE 1. Frozen complete-case packet yield (189 observations). Counts are unique within observations; timings are sums of concurrent-host arm wall times.

Receiver or set	Frames	Wall time (s)
Dire Wolf	2973	664.3
gr-satellites	1872	709.5
External union B	2998	—
Progressive P	4541	33888.0
Experimental I1, no codec	4703	26337.6
Experimental I2, codec/pool	4703	27212.6

In the signal-labelled complete subgroup, P recovers 136 packets versus 87 in B : 49 additions, zero losses, and a 56.32% net increase. Five of the 16 observations show at least one addition. These small, exploratory subgroup counts should accompany the overall metric rather than be promoted as a stronger independent trial.

I1 and I2 produce exactly identical observation-level packet sets in all 189 cases, totaling 4703. Therefore the current data show zero additional union packets from enabling I2 relative to I1. Relative to P, I2 adds 214 packets while missing 52, a net 162. P and I2 are complementary and neither observed set is a superset of the other. These comparisons do not isolate the causal value of individual components because the portfolios differ. Selecting their union after observing these data would be exploratory and would require a newly frozen evaluation policy.

The progressive arm averages 179.3 s per recording, compared with 3.5 s for Dire Wolf and 3.8 s for gr-satellites on this shared-host snapshot. Its aggregate wall-time ratio is approximately 51.0 times Dire Wolf and 47.8 times gr-satellites. Although P is faster than the recorded audio duration in aggregate, that fact does not establish streaming latency, deadline compliance, or isolated hardware efficiency. The extra recovery has a substantial computational cost.

A separate performance qualification on two previously used development recordings, each tested twice in paired order, measured total progressive wall time decreasing from 2199.86 to 1186.93 s (46.05%). Deterministic result projections agreed in all four pairs. This supports the specific optimization/build combination on those controls, not a mathematical zero-loss guarantee for every possible input or a 46% universal speedup. This small development comparison is distinct from the larger field-yield study.

VII. LIMITATIONS AND SUBMISSION REQUIREMENTS

A. VALIDITY AND GENERALIZATION

The present evidence is incomplete, single-mission, and retrospectively exposed. Stations, passes, dates, and repeated spacecraft content create dependence. A final analysis must use the registered dependence-aware procedure, include missingness and completion by subgroup, and state its resampling unit. Treating every packet as an independent Bernoulli trial would produce misleading uncertainty estimates. A finished cohort alone does not turn these results into an independent

validation.

The paper describes current implementation, not a causal proof that guarded transfer, blind fitting, or model pooling explains the observed gain. Matched ablations must retain common frontends, timing banks, input precision, resource policies, and acceptance checks while disabling one component at a time. Codec experiments require matched white, residual-only, codec, and pooled-codec controls with successful and rejected calibrations reported. The observed null I2/I1 packet result must be retained even if later experiments change it.

Dedicated no-transmission, out-of-band, wrong-profile, and controlled corrupt-packet tests are needed to bound false acceptance under the complete search policy. Representative added and lost packets should be independently traced back to waveform spans. Semantic consistency checks can strengthen evidence but cannot substitute for source authentication or become an undisclosed acceptance filter tuned on the test set.

B. IMPLEMENTATION VERSUS QUALIFIED COVERAGE

The generic Rust receiver contains separate complex-IQ BPSK, QPSK, and offset-QPSK paths; CSP version 1/2, selected CCSDS transfer formats, Reed–Solomon, and explicit sparse-matrix low-density parity-check facilities are also implemented. These paths have configuration and qualification boundaries and are not the receiver assessed in Table 1. This manuscript establishes no orbital QPSK/offset-QPSK gain, no universal CCSDS coding support, and no automatic unknown-modulation discovery. Post-FM mono OGG cannot be reinterpreted as phase-preserving IQ for those coherent paths.

Deployment qualification additionally requires documented supported inputs and profiles, reproducible installation, dependency/environment pinning, resource admission and restart tests, and independent review of the final results. The research web page accompanying this draft must distinguish available experimental code from production support or a hosted decoding service.

VIII. CONCLUSION

A resumable multi-hypothesis audio receiver can recover packets missed by a particular pair of established replay configurations. In a frozen interim set of 189 CANVAS observations, the progressive implementation yields 51.5% more per-observation unique UI packets than their union, at substantially greater computation cost and with four reference packets missed. This motivates a complementary offline recovery workflow rather than a universal replacement claim. The codec-enabled experimental arm adds no packets over its codec-disabled control in this snapshot. Completing and independently auditing the cohort, measuring component ablations and false acceptance, and evaluating genuinely separated mission/link data are necessary before stronger scientific or deployment claims.

DATA AND SOFTWARE AVAILABILITY

The local companion bundle contains frozen per-observation packet sets and counts, source/result identities, executable manifests, the replay profile, this LaTeX source, and checksums. The original OGG files remain in the existing research archive and are not duplicated in the manuscript bundle. No public repository, permanent dataset identifier, or public software release is claimed. Public artifact availability, dependency closure, and author-approved release arrangements must be completed before submission. The snapshot is derived from committed benchmark summaries; it is not a substitute for the campaign’s final raw-artifact validation.

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